

Chapter 1: Dynamics

Newton's Second Law

The acceleration of an object as produced by a net force is directly proportional to the magnitude of the net force, in the same direction as the net force, and inversely proportional to the mass of the object.

Formula: $F = ma$.

Frictionless Surfaces

If a surface is frictionless, there is no opposing force to motion along the surface.

Gravity

Near Earth's surface, gravity exerts a force $W = mg$, where g is approximately 9.8 m/s^2 .